

Programming for Data Science

# Medical Insurance Cost Analysis

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# Chapter 1: Introduction and Data Description

## 1.1 Introduction

Why do some people pay much more for health insurance than others? Insurance companies need to know this so they can set fair premiums. In this project we look at the medical bills of 1,338 people in the US and try to find out which things, like age, weight or smoking, make the bill go up.

We use three Python libraries: Pandas to load and group the data, NumPy to do the calculations, and Matplotlib to draw the charts.

## 1.2 Where the data comes from

We took the *Medical Cost Personal Datasets* from Kaggle ([kaggle.com/datasets/mirichoi0218/insurance](https://kaggle.com/datasets/mirichoi0218/insurance)). It is a single CSV file, `insurance.csv`. The data comes from the book *Machine Learning with R* and is based on US Census figures, so it is secondary data.

## 1.3 What is in the data

There are 1,338 rows (one per person) and 7 columns. Three columns are categorical and four are numerical.

| Column   | Type        | Meaning                                    |
|----------|-------------|--------------------------------------------|
| age      | Numerical   | Age of the person (18 to 64)               |
| sex      | Categorical | male or female                             |
| bmi      | Numerical   | Body mass index (30 or more means obese)   |
| children | Numerical   | Number of children covered (0 to 5)        |
| smoker   | Categorical | yes or no                                  |
| region   | Categorical | northeast, northwest, southeast, southwest |
| charges  | Numerical   | Medical cost billed by the insurer, in USD |

## 1.4 What we want to find out

- Is the data clean, and what does it look like?
- Do smokers, men or women, or people in some regions pay more?
- How much do age and BMI add to the bill?
- Which factor matters the most?

## Chapter 2: Data Analysis using Python

All the code is in `insurance_analysis.py`. For each part we show the code, what it printed, and what we learned from it.

### Part 2.1: Pandas

#### *Application 1: Loading and cleaning the data*

##### Code

```
def pandas_1_load_and_clean():
    df = pd.read_csv(DATA_FILE)
    print("Rows and columns:", df.shape)
    print("Missing values:", df.isnull().sum().sum())
    print("Duplicate rows:", df.duplicated().sum())

    df = df.drop_duplicates()
    print("Rows after removing duplicates:", len(df))

    print("\nSmokers:")
    print(df["smoker"].value_counts())
    print("\nSummary of numeric columns:")
    print(df.describe().round(1))
    return df
```

##### Result

```
Rows and columns: (1338, 7)
Missing values: 0
Duplicate rows: 1
Rows after removing duplicates: 1337

Smokers:
smoker
no      1063
yes      274
Name: count, dtype: int64

Summary of numeric columns:
      age    bmi  children  charges
count  1337.0  1337.0    1337.0   1337.0
mean    39.2   30.7        1.1  13279.1
std     14.0    6.1        1.2  12110.4
min     18.0   16.0        0.0   1121.9
25%     27.0   26.3        0.0   4746.3
50%     39.0   30.4        1.0   9386.2
75%     51.0   34.7        2.0  16657.7
max     64.0   53.1        5.0  63770.4
```

##### Inference

The data is in good shape. There are no missing values and only one duplicate row, which we removed, so we are left with 1,337 people. About one in five people smoke (274 out of 1,337). The average bill is \$13,279, but the median is only \$9,386. That gap tells us a few people have very large bills that pull the average up.

## Application 2: Comparing groups with groupby

### Code

```
def pandas_2_group_by(df):  
    print("Average charges by smoker:")  
    print(df.groupby("smoker")["charges"].mean().round(0))  
  
    print("\nAverage charges by sex:")  
    print(df.groupby("sex")["charges"].mean().round(0))  
  
    print("\nAverage charges by region:")  
    print(df.groupby("region")["charges"].mean().round(0))
```

### Result

```
Average charges by smoker:  
smoker  
no      8441.0  
yes     32050.0  
Name: charges, dtype: float64  
  
Average charges by sex:  
sex  
female   12570.0  
male     13975.0  
Name: charges, dtype: float64  
  
Average charges by region:  
region  
northeast   13406.0  
northwest   12451.0  
southeast   14735.0  
southwest   12347.0  
Name: charges, dtype: float64
```

### Inference

Smoking makes a huge difference. Smokers pay about \$32,050 on average, while non-smokers pay about \$8,441, which is almost four times less. Men pay a little more than women, and the southeast is a bit more expensive than other regions, but these gaps are small compared with the smoking gap.

### Application 3: Creating new columns with `pd.cut`

#### Code

```
def pandas_3_new_columns(df):
    df["age_group"] = pd.cut(df["age"], bins=[17, 29, 39, 49, 64],
                             labels=["18-29", "30-39", "40-49", "50-64"])
    df["obese"] = df["bmi"] >= 30

    print("Number of obese people:", df["obese"].sum())
    print("\nAverage charges by age group and smoker:")
    print(df.pivot_table(values="charges", index="age_group",
                          columns="smoker", observed=True).round(0))
    print("\nAverage charges by obese and smoker:")
    print(df.pivot_table(values="charges", index="obese",
                          columns="smoker").round(0))
    return df
```

#### Result

```
Number of obese people: 706

Average charges by age group and smoker:
smoker      no      yes
age_group
18-29      4427.0  27518.0
30-39      6337.0  30271.0
40-49      9183.0  32655.0
50-64     13431.0  38748.0

Average charges by obese and smoker:
smoker      no      yes
obese
False      7977.0  21363.0
True       8856.0  41558.0
```

#### Inference

Bills go up with age for everyone. A non-smoker aged 50-64 pays about three times as much as one aged 18-29. More than half of the people (706) are obese. Being obese hardly changes the bill for non-smokers (\$7,977 vs \$8,856), but for smokers it almost doubles it (\$21,363 vs \$41,558). So smokers who are also obese are the most expensive group.

## Part 2.2: NumPy

### Application 1: Basic statistics and outliers

#### Code

```
def numpy_1_statistics(df):
    charges = df["charges"].to_numpy()

    print("Mean: ", np.mean(charges).round(2))
    print("Median:", np.median(charges).round(2))
    print("Std: ", np.std(charges).round(2))

    q1, q3 = np.percentile(charges, [25, 75])
    limit = q3 + 1.5 * (q3 - q1)
    outliers = charges[charges > limit]
    print("\nOutlier limit:", limit.round(2))
    print("Number of outliers:", len(outliers))
```

#### Result

```
Mean:    13279.12
Median:  9386.16
Std:     12105.83

Outlier limit: 34524.78
Number of outliers: 139
```

#### Inference

The mean (\$13,279) is much higher than the median (\$9,386), and the standard deviation is about \$12,100, so the bills are spread out a lot. Using the usual rule ( $Q3 + 1.5 \times IQR$ ), 139 people are outliers with bills above \$34,525. We kept them, because these are real people with high costs, not mistakes in the data.

## Application 2: Correlation with `np.corrcoef`

### Code

```
def numpy_2_correlation(df):
    smoker = np.where(df["smoker"] == "yes", 1, 0)
    data = np.array([df["age"], df["bmi"], df["children"], smoker, df["charges"]])
    names = ["age", "bmi", "children", "smoker", "charges"]

    corr = np.corrcoef(data)
    print("Correlation with charges:")
    for name, value in zip(names[:-1], corr[:-1, -1]):
        print(f" {name:<9} {value:.2f}")
    return names, corr
```

### Result

```
Correlation with charges:
age      0.30
bmi      0.20
children 0.07
smoker   0.79
```

### Inference

We turned smoker into 1 (yes) and 0 (no) so we could measure it. Smoking has by far the strongest link with charges (0.79). Age comes next (0.30), then BMI (0.20). The number of children has almost no link (0.07).

### Application 3: A simple prediction model

#### Code

```
def numpy_3_regression(df):
    smoker = np.where(df["smoker"] == "yes", 1, 0)
    X = np.column_stack([np.ones(len(df)), df["age"], df["bmi"], smoker])
    y = df["charges"].to_numpy()

    b = np.linalg.lstsq(X, y, rcond=None)[0]
    predicted = X @ b
    r2 = 1 - np.sum((y - predicted) ** 2) / np.sum((y - y.mean()) ** 2)

    print("Intercept:", b[0].round(0))
    print("Per year of age:", b[1].round(0))
    print("Per BMI point:", b[2].round(0))
    print("For smoking:", b[3].round(0))
    print("R squared:", r2.round(3))
```

#### Result

```
Intercept: -11672.0
Per year of age: 259.0
Per BMI point: 323.0
For smoking: 23822.0
R squared: 0.747
```

#### Inference

We fitted a straight-line model using age, BMI and smoking. It says each extra year of age adds about \$259, each BMI point adds about \$323, and smoking adds about \$23,800. These three columns alone explain about 75% of the differences in charges ( $R^2 = 0.747$ ), which is quite good for such a simple model.

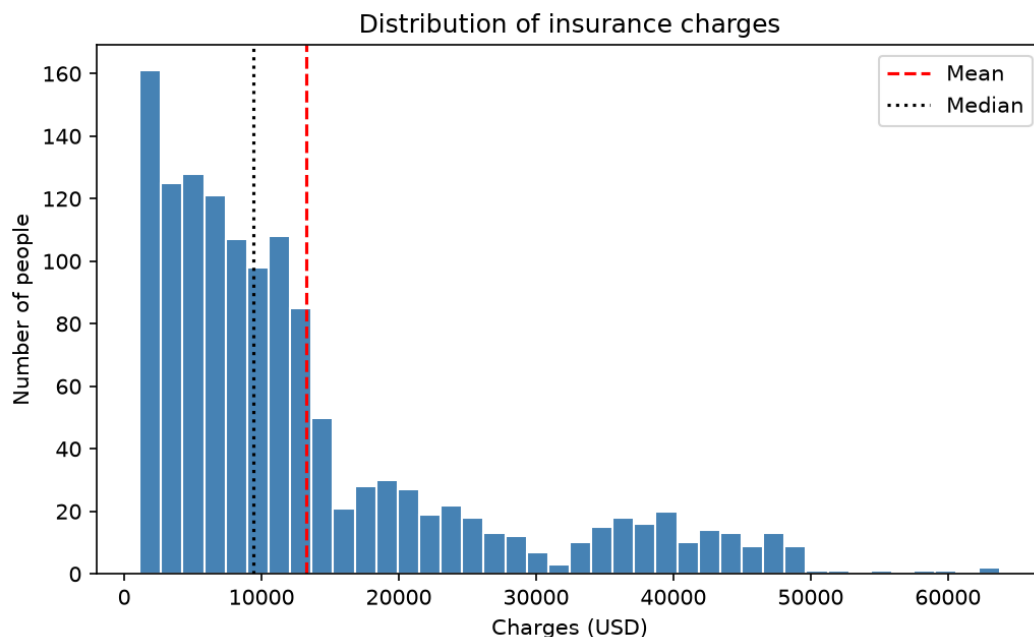
## Part 2.3: Matplotlib

### Application 1: Histogram of charges

#### Code

```
def matplotlib_1_histogram(df):  
    plt.figure(figsize=(8, 4.5))  
    plt.hist(df["charges"], bins=40, color="steelblue", edgecolor="white")  
    plt.axvline(df["charges"].mean(), color="red", linestyle="--", label="Mean")  
    plt.axvline(df["charges"].median(), color="black", linestyle=":", label="Median")  
    plt.title("Distribution of insurance charges")  
    plt.xlabel("Charges (USD)")  
    plt.ylabel("Number of people")  
    plt.legend()  
    plt.savefig(FIG_DIR / "mpl_1_histogram.png", dpi=150, bbox_inches="tight")  
    plt.close()
```

#### Result



#### Inference

Most people pay less than \$15,000, but the chart has a long tail on the right that goes past \$60,000. The mean line sits to the right of the median line because of those few big bills.

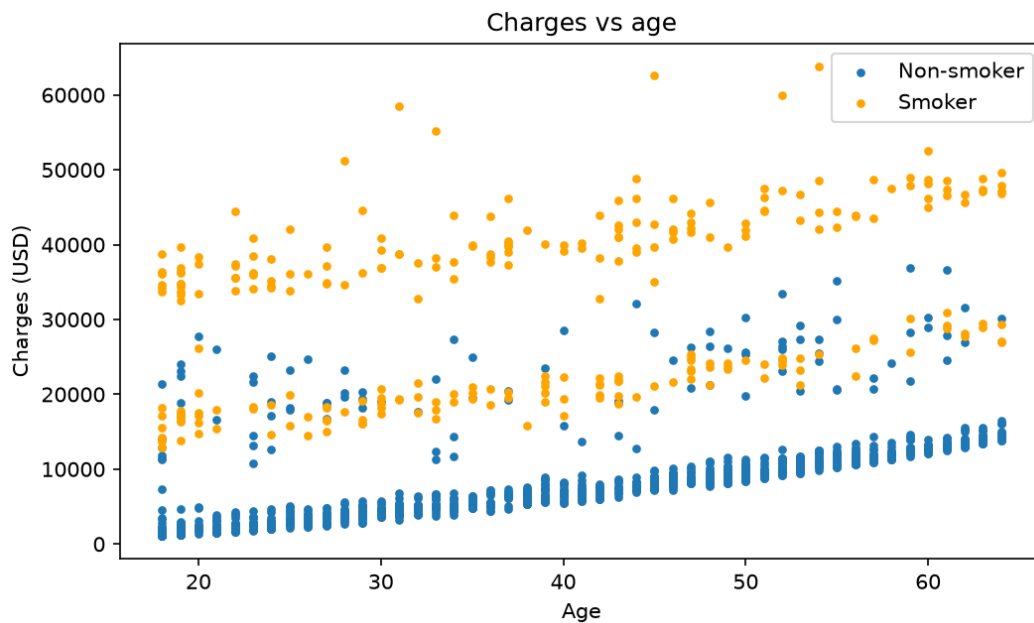
## Application 2: Scatter plot of charges vs age

### Code

```
def matplotlib_2_scatter(df):
    smokers = df[df["smoker"] == "yes"]
    non_smokers = df[df["smoker"] == "no"]

    plt.figure(figsize=(8, 4.5))
    plt.scatter(non_smokers["age"], non_smokers["charges"], s=10, label="Non-smoker")
    plt.scatter(smokers["age"], smokers["charges"], s=10, color="orange", label="Smoker")
    plt.title("Charges vs age")
    plt.xlabel("Age")
    plt.ylabel("Charges (USD)")
    plt.legend()
    plt.savefig(FIG_DIR / "mpl_2_scatter.png", dpi=150, bbox_inches="tight")
    plt.close()
```

### Result



### Inference

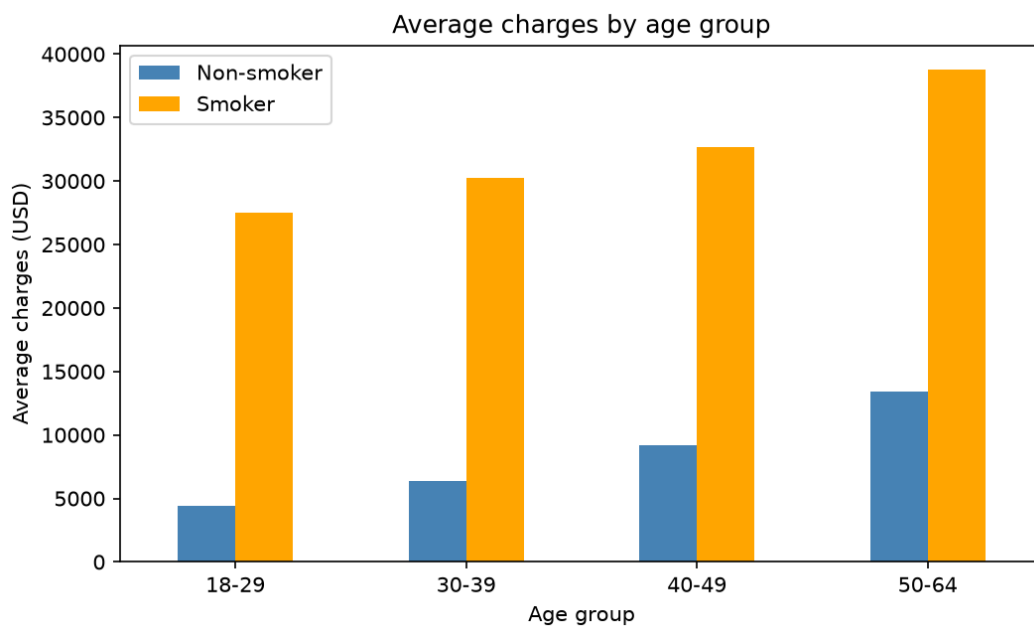
This is the most useful chart. The dots form three bands. The bottom band is almost all non-smokers, and the top band is only smokers. In every band, the bill rises steadily as people get older.

### Application 3: Bar chart by age group

#### Code

```
def matplotlib_3_bar_chart(df):  
    table = df.pivot_table(values="charges", index="age_group",  
                           columns="smoker", observed=True)  
    table.columns = ["Non-smoker", "Smoker"]  
  
    table.plot(kind="bar", figsize=(8, 4.5), color=["steelblue", "orange"])  
    plt.title("Average charges by age group")  
    plt.xlabel("Age group")  
    plt.ylabel("Average charges (USD)")  
    plt.xticks(rotation=0)  
    plt.savefig(FIG_DIR / "mpl_3_bar_chart.png", dpi=150, bbox_inches="tight")  
    plt.close()
```

#### Result



#### Inference

In every age group the orange bar (smokers) is far taller than the blue bar (non-smokers). A young smoker pays more than twice what the oldest non-smokers pay on average.

# Chapter 3: Conclusion

## What we found

- Smoking is the biggest reason for high medical bills. Smokers pay almost four times more than non-smokers.
- Older people pay more. Each year of age adds about \$259 to the bill.
- Being obese mostly hurts smokers. It nearly doubles their bill but barely changes a non-smoker's.
- Sex, region and number of children make only a small difference.

## The story

If we had to guess someone's medical bill, the first question to ask is "Do you smoke?" The next questions would be their age and their weight. Put simply, a young non-smoker is cheap to insure, and an older, obese smoker is very expensive. Pandas helped us spot these groups, NumPy measured how big the effects are, and the Matplotlib charts made the pattern easy to see.

## Suggestions for the insurer

- Price policies mainly on smoking, age and BMI.
- Offer programmes to help people quit smoking and lose weight. This is where the biggest savings are.
- Do not charge more based on sex or region.

## Limitations

The data is simulated and has only 1,338 people. It has no medical history or income, so our results show links, not proof of cause.